

RAKSHITH SRIRAMAN KRISHNARAJ

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Education

University of Wisconsin Madison

Madison, WI

Master of Science in Computer Science GPA: 4.0

Aug 2025 – May 2027

- **Relevant coursework:** Machine Learning, Natural Language Processing, Computer Vision, Trustworthy AI, Big Data Systems, Reinforcement Learning, Foundation Models

MIT Campus, Anna University

Chennai, India

Bachelor of Engineering in Electronics and Instrumentation, GPA: 3.87

Aug 2019 – May 2023

University of Tours

Tours, France

Student Exchange Program, GPA: 3.92

Aug 2021 – Feb 2022

Skills

Languages: Python, C++, Java, HTML, JavaScript, CSS, SQL, MATLAB

Frameworks/Tools: AWS, GCP, ReactJS, NodeJS, PyTorch, TensorFlow, Scikit, gRPC, Git, Azure, PostgreSQL, MySQL, Cassandra, HBase, BigQuery, Milvus, Hadoop, Spark, MapReduce, Kafka, Elasticsearch, Android Dev, Heroku

Experience

Accenture

Chennai, India

Data Engineer

September 2023 – August 2025

- Built a distributed RAG ingestion framework to process high-volume financial data using Python (**FastAPI, Flask**), OpenAI APIs, **Milvus, PostgreSQL and PowerBI**, reducing pipeline runtime from 5 days to 24 hours (80% improvement) while achieving 95% query accuracy and supporting 500+ daily real-time queries.
- Architected a multi-agent RAG query system with intent classification, query decomposition, and hybrid retrieval (Milvus search + PostgreSQL) to serve 3,000+ company financial profiles across structured and unstructured data sources.

National Institute of Ocean Technology

Chennai, India

Data Engineering Intern

July 2022 – June 2023

- Engineered a Model Predictive Control algorithm for the Ocean Thermal Energy Conversion (OTEC) process using **MATLAB**, optimizing turbine power output under thermal and pressure constraints, reducing steady-state error by 25% and improving transient response time by 18%.
- Trained an **LSTM** model to reject disturbance in an OC-OTEC plant, **improving** system stability by **30%**.

Carjoz

Remote

Software Development Engineering Intern

September 2021 – January 2022

- Implemented an **Android** app for workers to efficiently enter and manage customer data supporting 200+ daily records.
- Utilized **Firebase** to optimize a cloud database, ensuring data integrity and accessibility for 50+ active users.

Publications

Decentralized Control Strategies with Predictive Disturbance Rejection for OC-OTEC Plant in Lakshadweep Using Deep Learning

JWPE 2024

- Engineered a deep learning driven decentralized control system for a lab-scale OC-OTEC plant, achieving 98.6% Sea Surface Temperature (SST) prediction accuracy and reducing disturbance-induced power fluctuations by over 30%.

Multiple Hybrid AI Model-based Optimal Charging Profile of Lithium-ion Battery for SoH Enhancement

IEEE 2023

- Designed multiple hybrid AI models for 18650 Li-ion batteries, achieving up to 30% faster charging and 12% lower temperature rise, significantly improving battery lifespan and reliability for EV applications.

Data-driven Model Development and Validation for Laboratory Scale OC-OTEC Plant

IEEE 2022

- Formulated linear data-driven SISO and MIMO models for OC-OTEC processes, improving system efficiency by 50% using real-time data from NIOT's laboratory setup.

Projects

Distributed Weather Streaming System (Kafka, gRPC, Docker)

Fall 2025

- Engineered a Kafka-based real-time streaming pipeline processing 10 messages/sec across 4 partitions, using Protobuf serialization and offset checkpointing to achieve fault-tolerant, exactly-once processing with crash recovery.

AI-based BMS for SoH Enhancement

Spring 2023

- Engineered multiple AI models using NASA's dataset and optimized charging profiles with particle swarm optimization, reducing charging time by 20% and limiting temperature rise in Li-ion batteries.

COVID-19 Prediction using Chest X-rays

Summer 2021

- Designed an advanced convolutional neural network to classify COVID-19 infections from chest X-ray images, reaching 95% diagnostic accuracy and processing 1000+ images.

Certifications

Machine Learning (Coursera) — Android Developer (Udemy) — Java Certification (Cambridge) — Medical Image Analysis (IET)